

FastHenry2Object.Run

boolean FastHenry2Object.**Run**(**BSTR** *commandLine*)

Return Value

True if FastHenry2 was able to start a simulation, false otherwise.

Notes:

A reason why FastHenry2 could not be able to start a simulation is that a simulation is already running. You can try to [stop](#) FastHenry2 before retrying and / or [poll FastHenry2 status](#).

Parameters

commandLine

A string containing a FastHenry2 command line, as would be given from a shell console.

Example:

```
FastHenry2Object.Run("broken.inp -O")
```

Remarks

Runs a simulation.

See Also

[FastHenry2 Automation Overview](#) | [IsRunning Property](#) | [Stop Method](#)

FastHenry2Object.Stop

void FastHenry2Object.**Stop**()

Return Value

None

Parameters

None

Remarks

Stops the currently running simulation.

Notes:

It could take some time before FastHenry2 actually stops, depending on the operation currently under execution. You can poll FastHenry2 status checking the [IsRunning](#) property or you can set a callback upon FastHenry2 end of operation using the [SetEndCallback](#) method.

See Also

[FastHenry2 Automation Overview](#) | [IsRunning Property](#) | [SetEndCallback Method](#)

FastHenry2Object.Quit

boolean FastHenry2Object.**Quit**()

Return Value

If true, FastHenry2 was able to quit. If false, FastHenry2 couldn't quit (possibly because there's a simulation on going; try to [stop](#) it before quitting).

Parameters

None

Remarks

Quits FastHenry2.

See Also

[FastHenry2 Automation Overview](#) | [Stop Method](#)

FastHenry2Object.ShowWindow

void FastHenry2Object.**ShowWindow()**

Return Value

None

Parameters

None

Remarks

Shows FastHenry2 console window, which is hidden by default when the FastHenry2 object is created via automation.

Notes:

Once the console becomes visible, you cannot hid it again.

See Also

[FastHenry2 Automation Overview](#)

FastHenry2Object.GetInductance

VARIANT FastHenry2Object.**GetInductance()**

Return Value

The method returns a 3-dimensional safe-array encapsulated in a variant data.

The first dimension identifies the matrix solved for a particular frequency. To know the frequency, you must query the frequency array via the [GetFrequencies Method](#). The index of both arrays is the same, e.g. the second matrix in the inductance array is calculated at the frequency in the second position of the frequency array.

The second and third dimensions correspond to the row and column index respectively.

If the number of columns equals the number of rows of the array, the array contains the inductance matrix as calculated by FastHenry2, in Henry.

If the number of columns of the array is not equal to the number of rows, the array contains the imaginary part of the rectangular admittance matrix (this case happens when option '-x' is used running FastHenry2, see FastHenry user's manual).

Notes:

The array dimensions can be queried from the safe array itself, see the 'safe array' topic in the MS Automation documentation.

Parameters

None

Remarks

Gets the inductance or the imaginary part of the admittance values calculated by FastHenry2, result of the

simulation.

See Also

[FastHenry2 Automation Overview](#) | [GetColPortNames Method](#) | [GetRowPortNames Method](#) | [GetResistance Method](#) | [GetFrequencies Method](#)

FastHenry2Object.GetResistance

VARIANT FastHenry2Object.GetResistance()

Return Value

The method returns a 3-dimensional safe-array encapsulated in a variant data.

The first dimension identifies the matrix solved for a particular frequency. To know the frequency, you must query the frequency array via the [GetFrequencies Method](#). The index of both arrays is the same, e.g. the second matrix in the resistance array is calculated at the frequency in the second position of the frequency array.

The second and third dimensions correspond to the row and column index respectively.

If the number of columns equals the number of rows of the array, the array contains the resistance matrix as calculated by FastHenry2, in Ohms.

If the number of columns of the array is not equal to the number of rows, the array contains the real part of the rectangular admittance matrix (this case happens when option '-x' is used running FastHenry2, see FastHenry user's manual).

Notes:

The array dimensions can be queried from the safe array itself, see the 'safe array' topic in the MS Automation documentation.

Parameters

None

Remarks

Gets the resistance or the real part of the admittance values calculated by FastHenry2, result of the simulation.

See Also

[FastHenry2 Automation Overview](#) | [GetColPortNames Method](#) | [GetRowPortNames Method](#) | [GetInductance Method](#) | [GetFrequencies Method](#)

FastHenry2Object.GetFrequencies

VARIANT FastHenry2Object.GetFrequencies()

Return Value

The method returns a 1-dimensional safe-array encapsulated in a variant data. The array contains the frequencies at which the [resistance](#), [inductance](#) or [admittance](#) matrices are calculated.

Parameters

None

Remarks

Gets the list of frequency points at which FastHenry2's results are calculated.

See Also

[FastHenry2 Automation Overview](#) | [GetColPortNames Method](#) | [GetRowPortNames Method](#) | [GetResistance Method](#) | [GetInductance Method](#)

FastHenry2Object.GetRowPortNames

VARIANT FastHenry2Object.**GetRowPortNames()**

Return Value

The method returns a 1-dimensional safe-array encapsulated in a variant data. The array contains the names of the ports to which the resistance, inductance, and admittance matrices rows refer.

Notes:

The array dimensions can be queried from the safe array itself, see the 'safe array' topic in the MS Automation documentation.

Parameters

None

Remarks

Gets port names for the rows of the resistance, inductance, and admittance matrices.

See Also

[FastHenry2 Automation Overview](#) | [GetColPortNames Method](#) | [GetResistance Method](#) | [GetInductance Method](#)

FastHenry2Object.GetColPortNames

VARIANT FastHenry2Object.**GetColPortNames()**

Return Value

The method returns a 1-dimensional safe-array encapsulated in a variant data. The array contains the names of the ports to which the resistance, inductance, and admittance matrices columns refer.

Notes:

The array dimensions can be queried from the safe array itself, see the 'safe array' topic in the MS Automation documentation.

Parameters

None

Remarks

Gets port names for the columns of the resistance, inductance, and admittance matrices.

See Also

[FastHenry2 Automation Overview](#) | [GetRowPortNames Method](#) | [GetResistance Method](#) | [GetInductance Method](#)

FastHenry2Object.SetEndCallback

boolean FastHenry2Object.**SetEndCallback(IDispatch* *callback*, BSTR *cbName*)**

Return Value

True if a callback function was set. False if the callback function is reset to default (no callback).

Parameters

callback

Pointer to the client's automation object exposing the callback function. If NULL is passed as argument, the callback will be removed.

cbName

Name of the callback function.

The callback function must have the following prototype:

```
void cbName()
```

Remarks

Call this function to set a callback upon FastHenry2 simulation end. That is, when FastHenry2 has finished the simulation and results are available, the *cbName* function will be called to notify the client.

See Also

[FastHenry2 Automation Overview](#) | [IsRunning Property](#) | [SetLogCallback Method](#)

FastHenry2Object.SetLogCallback

```
boolean FastHenry2Object.SetLogCallback(IDispatch* callback, BSTR cbName)
```

Return Value

True if a callback function was set. False if the callback function is reset to default (no callback).

Parameters

callback

Pointer to the client's automation object exposing the callback function. If NULL is passed as argument, the callback will be removed.

cbName

Name of the callback function.

The callback function must have the following prototype:

```
void cbName(BSTR logString, long color)
```

logString

The string containing the log text passed back to the client.

color

The color of the string. The color will be black (red 255, green 255, blue 255) for normal messages and red (red 255, green 0, blue 0) for error messages.

Remarks

Call this function to set a callback upon FastHenry2 log output. That is, when FastHenry2 has something to log (everything that would normally be visible in FastHenry2's console window), the *cbName* function will be called to notify the client and provide him the output string. This function is especially useful if FastHenry2's console window is hidden.

See Also

[FastHenry2 Automation Overview](#) | [ShowWindow Method](#) | [SetEndCallback Method](#)

FastHenry2Object.IsRunning

```
boolean FastHenry2Object.IsRunning()
```

Return Value

True if there's a simulation running. False if FastHenry2 is idle.

Parameters

None

Remarks

Tests if FastHenry2 is running a simulation.

See Also

[FastHenry2 Automation Overview](#) | [SetEndCallback Method](#) | [Stop Method](#)